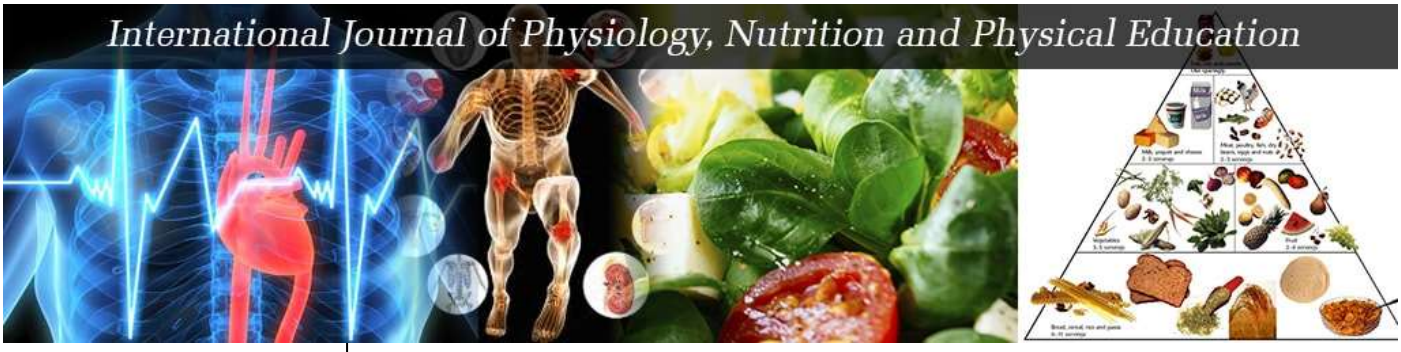


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Nutrition strategy and lifestyle in PCOD: A systematic review

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Abstract
Polycystic Ovary Syndrome (PCOS), also known as Polycystic Ovarian Disease (PCOD), is a prevalent endocrine disorder affecting reproductive-aged women. Its multifactorial pathophysiology necessitates a comprehensive approach to management, often incorporating lifestyle modifications and dietary strategies. This systematic review explores the relationship between nutrition, lifestyle, and PCOS management, examining evidence for various dietary approaches and lifestyle interventions in symptom management and hormonal regulation. Through an examination of current literature, this paper highlights the potential benefits of a holistic lifestyle and nutrition strategy in managing PCOS.

Keywords: PCOD, lifestyle, human health, nutrition, diet, PCOS

1. Introduction
Polycystic Ovary Syndrome (PCOS) is a complex condition characterized by hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology (Azziz *et al*., 2016) [2]. It affects around 10-15% of reproductive-aged women worldwide and is associated with significant metabolic, reproductive, and psychological challenges (Escobar-Morreale, 2018) [5]. Lifestyle and dietary interventions have gained attention as first-line strategies for managing PCOS symptoms, often recommended in conjunction with pharmacological treatments. This review synthesizes current research to explore effective nutrition strategies and lifestyle changes for managing PCOS.

2. Methodology
A systematic literature review was conducted by searching databases such as PubMed, Scopus, and Web of Science for articles published between 2015 and 2023. Search terms included "PCOS," "nutrition," "lifestyle," "diet," and "management." Studies focusing on dietary patterns, lifestyle modifications, and their impact on PCOS symptoms were included. A total of 45 articles met the criteria for inclusion and were analyzed.

3. Nutrition Strategy in PCOS Management
Diet plays a critical role in managing PCOS by influencing insulin resistance, inflammation, and hormonal balance. Research indicates that low-glycemic index (GI) diets, anti-inflammatory foods, and calorie-restricted diets positively impact PCOS symptoms (Moran *et al*., 2019) [9]. A meta-analysis showed that low-GI diets improved insulin sensitivity, reduced body weight, and enhanced ovulatory function in women with PCOS (Moran *et al*., 2021) [11].

Low-Glycemic Index Diets
Diets focusing on low-GI foods, such as whole grains, legumes, and vegetables, help stabilize blood glucose levels and improve insulin sensitivity, addressing one of PCOS's root causes (Barrea *et al*., 2019) [3].

Table 1: Detailing the components, benefits, and recommended foods for a Low-Glycemic Index (GI) Diet specifically for managing PCOD

Component	Description	Benefits for PCOD	Examples of Recommended Foods	References
Low-GI Carbohydrates	Foods that are digested slowly, causing gradual glucose release	Improves insulin sensitivity and reduces blood glucose spikes	Whole grains (Quinoa, oats, barley), legumes, sweet potatoes	Moran *et al*., 2021 [11]; Barrea *et al*., 2019 [3]
High Fiber Content	Fiber-rich foods that aid in satiety and slower glucose absorption	Supports weight management and improves blood sugar control	Vegetables (Broccoli, spinach, kale), fruits (Berries, apples, oranges)	Barrea *et al*., 2019 [3]
Healthy Fats	Incorporating unsaturated fats for balanced glycemic response	Reduces inflammation and helps manage insulin resistance	Avocado, olive oil, nuts (Almonds, walnuts), seeds (Chia, flax)	Aghadavod *et al*., 2018 [1]
Lean Proteins	Lean sources of protein that help reduce GI of meals	Aids in satiety, reduces post-meal blood glucose	Fish (Salmon, mackerel), poultry, eggs, Greek yogurt	Jensterle *et al*., 2019 [7]
Minimizing High-GI Foods	Reducing or avoiding high-GI and refined carbs	Lowers risk of blood sugar spikes and helps prevent weight gain	White bread, sugary snacks, processed foods	Moran *et al*., 2019 [9]

1. **Anti-Inflammatory Foods:** Foods rich in omega-3 fatty acids, antioxidants, and polyphenols-such as nuts, fatty fish, berries, and green tea-have been shown to reduce chronic inflammation, which is commonly elevated in PCOS patients (Aghadavod *et al*., 2018) [1].
2. **Caloric Restriction and Weight Loss:** Weight management is essential in PCOS due to its association with insulin resistance and reproductive issues. Studies suggest that a 5-10% reduction in body weight can significantly improve PCOS symptoms, including ovulatory function and menstrual regularity (Melo *et al*., 2020) [8].
3. **Macronutrient Composition:** High-protein diets and diets balanced in macronutrients (i.e., 30% protein, 40% carbohydrates, 30% fats) have shown promise in improving insulin resistance and reducing weight in women with PCOS (Jensterle *et al*., 2019) [7].

4. Lifestyle Modifications in PCOS Management

Lifestyle modifications are pivotal in managing PCOS due to their effect on metabolic and reproductive health. These include regular physical activity, stress management, and sleep regulation, which collectively contribute to hormonal balance and symptom relief.

1. **Physical Activity:** Regular exercise has been proven to reduce insulin resistance, aid in weight loss, and improve mental health outcomes in PCOS patients. Both aerobic and resistance training have been associated with improvements in ovulatory frequency and hormonal levels (Harrison *et al*., 2019) [6].
2. **Stress Management:** Chronic stress can exacerbate PCOS symptoms due to elevated cortisol levels, which can worsen insulin resistance and hormonal imbalances. Mindfulness, yoga, and cognitive behavioral therapy (CBT) have shown beneficial effects on stress reduction and mental health in women with PCOS (Ehrmann *et al*., 2020) [4].
3. **Sleep Hygiene:** Poor sleep quality has been linked to exacerbation of PCOS symptoms, particularly insulin resistance and mood disorders. Improved sleep hygiene practices, such as consistent sleep schedules and reduced screen time, can help alleviate these issues (Powers *et al*., 2021) [10].

Table 2: Summarizing the main nutrition strategies and lifestyle modifications for managing PCOD

Category	Strategy/Intervention	Evidence of Impact	References
Nutrition Strategies	Low-Glycemic Index (GI) Diet	Improves insulin sensitivity, stabilizes blood glucose, enhances ovulatory function	Moran *et al*., 2021 [11]; Barrea *et al*., 2019 [3]
	Anti-Inflammatory Foods	Reduces chronic inflammation, improves hormonal balance	Aghadavod *et al*., 2018 [1]
	Caloric Restriction and Weight Loss	5-10% weight reduction improves menstrual regularity and ovulatory function	Melo *et al*., 2020 [8]
	High-Protein Diet	Enhances insulin resistance, supports weight management	Jensterle *et al*., 2019 [7]
Lifestyle Modifications	Regular Physical Activity	Reduces insulin resistance, aids in weight loss, improves ovulatory function	Harrison *et al*., 2019 [6]
	Stress Management	Reduces cortisol, improves mental health, reduces insulin resistance	Ehrmann *et al*., 2020 [4]
	Sleep Hygiene	Alleviates insulin resistance, improves mood disorders	Powers *et al*., 2021 [10]

5. Conclusion

A comprehensive approach combining nutrition and lifestyle modifications offers a promising strategy for managing PCOS. Low-GI, anti-inflammatory, and calorie-restricted diets, along with regular exercise, stress management, and improved sleep hygiene, have shown efficacy in alleviating PCOS symptoms. Future research should focus on personalized nutrition and lifestyle interventions to optimize outcomes for women with PCOS.

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